Technische Universität Dresden

CIDS ZIH

Machine Learning II

Digitization and Data Analytics

Structure of the Lecture Series

Lecture 1: Introduction

Lecture 2: Data Preprocessing

Lecture 3: Research Data Management

Lecture 4: Machine Learning I - Unsupervised Learning

Lecture 5: Machine Learning II - Supervised Learning

Lecture 6: Machine Learning III

Lecture 7: Data Analytics and Statistics

Lecture 8: Deep Learning I

Lecture 9: Deep Learning II

Lecture 10: Distributed Computing I

Lecture 11: Distributed Computing II

Lecture 12: Consequences

Last Lecture

Introduction Machine Learning
- Unsupervised Learning
- Supervised Learning

Similarities and Distances
- Clustering
- Partitioning
- Hierarchical clustering
- Density-based
- Assessment of clustering results
- Applications

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
4.9	3.1	1.5	0.1
5.1	3.8	1.5	0.3
4.7	3.2	1.6	0.2
5.1	3.4	1.5	0.2
5.0	3.3	1.4	0.2
5.2	2.7	3.9	1.4
5.7	2.6	3.5	1.0
5.7	2.8	4.1	1.3
6.0	2.2	5.0	1.5

Sepal.Length	Sepal.Width	Petal.Length	Petal.Widt	Species
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5.1	3.8	1.5	.3	setosa
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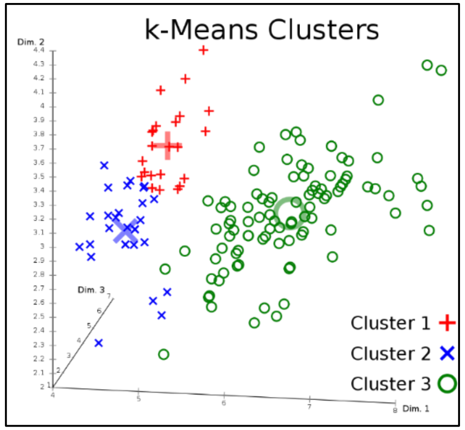
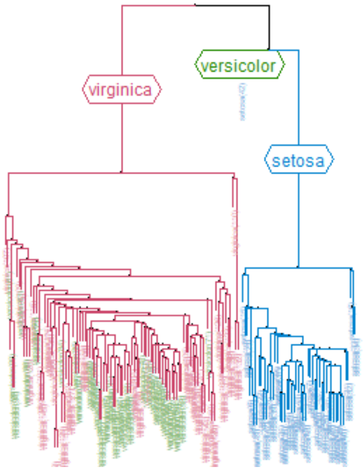
5.1	3.4	1.5	).2	setosa
5.0	3.3	1.4	).2	setosa
5.2	2.7	3.9	.4	versicolor
5.7	2.6	3.5	.0	versicolor
5.7	2.8	4.1	1 3	versicolor
6.0	2.2	5.0	1 5	virginica



Cluster Dendogram for iris
distance euclidean
method single



Structure of this Lecture

1. Supervised Learning – Training and Learning

2. Classification

    2.1 Decision trees

    2.2 Bayes Classification Methods

- 3. Model Evaluation and Selection

Recap Supervised Learning (Predictive Analysis)

Labeled Data

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4.9	3.1	1.5	0.1	setosa
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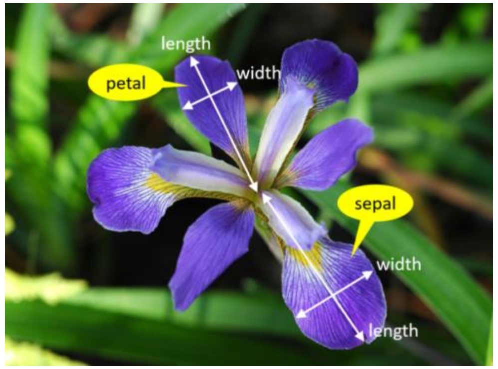
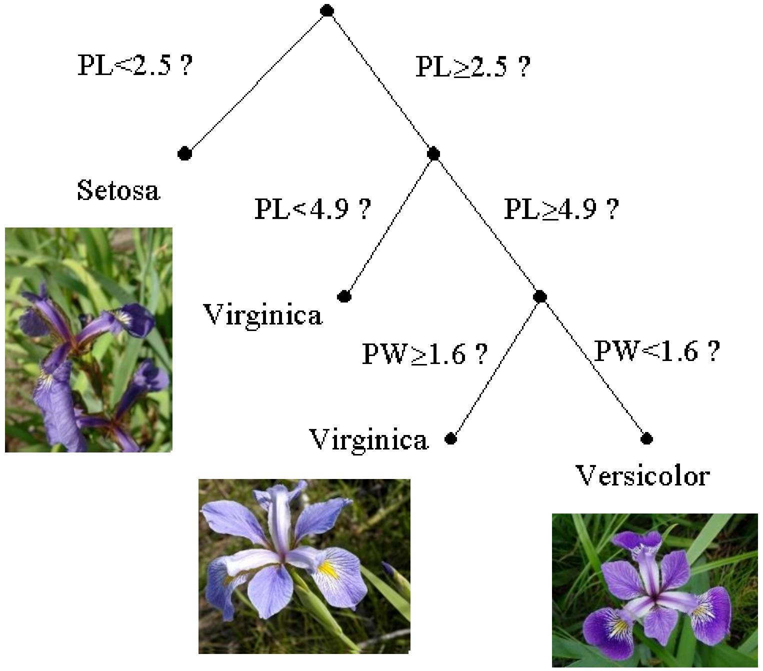
Courtesy of Iryna Okhrin



https://www.integratedots.com/determine-number-of-iris-species-with-k-means/

Goal: Classify unknown data



PL: Petal.Length,

PW: Petal.Width

***Iris* flower data set,** Ronald Fisher, *The use of multiple measurements in taxonomic problems,* 1936

Recap Supervised Learning - Classification

Given

- $D = \{(\vec{x}_j, \vec{y}_j)\}_{j=1}^{N}$ **training set** of **labeled** input-output pairs
- N **training examples**

supervised

$(\vec{x}, \vec{y})$ input-output pairs

x_i: **feature** (attribute, covariate, predictors)

y_i: **label** (response variable, target)

$\vec{x}$: **feature vector**

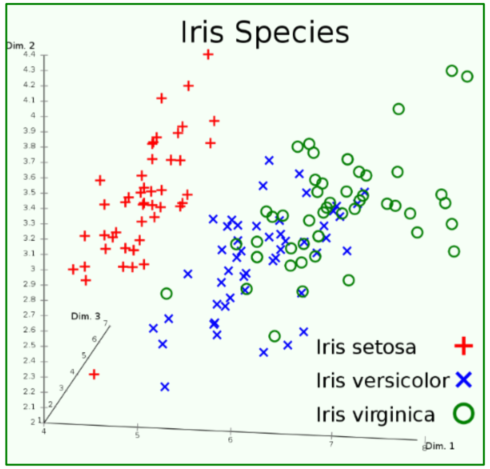
$\vec{y}$: **output vector**

Goal

- Learn mapping f: $\vec{x} \rightarrow \vec{y}$

Recap Supervised Learning - Classification

- **Informally:**
  - Describes properties of future data
  - Question „What could happen?“
  - Goal: Predicting future
    - E.g. forecasting, scoring, **classification of unknown data**
- **Based on formal description:**
  - **learn** (fit) a **model** that **relates** the **label** (response) to the **features** (predictors)
  - accurately predict the label for future observations
  - better understanding the relationship between the labels and the features
- **Probabilistic view:**
  - fitting a conditional model, $p(y|x)$, which specifies (a distribution over) outputs given inputs



Supervised Learning – Example Spam-filtering

Given:

X	insurance	learning	the	dating	medicament	**spam?**
$\vec{x}_1$	1	0	1	0	0	$y_1=1$
$\vec{x}_2$	0	1	1	0	0	$y_2=0$
$\vec{x}_3$	0	0	0	0	1	$y_3=1$

Instance space $\mathbf{x} \in \boldsymbol{X}$ ($|X| = n$ data points), describing emails

•Binary or real-valued feature vector $\vec{\boldsymbol{x}}$ (word occurrences, further information about document)

•***d*** features (*d*~100,000)

Class $\boldsymbol{y} \in \boldsymbol{Y}, \boldsymbol{Y} = \{\mathbf{0}, \mathbf{1}\}$**,**

$$\boldsymbol{y} = \begin{cases} \mathbf{1} & \mathbf{Spam} \\ \mathbf{0} & \mathbf{Non-Spam} \end{cases}$$

Goal: Estimate a function ***f(x)*** so that ***y = f(x)***

Training and Learning – Features and Labels

Prediction

- **Estimate** a function $f\colon \vec{x} \rightarrow \vec{y}$

Where $\vec{y}$ can be:

- **Categorical: Classification**
- **Real number:** Regression
- **Complex object**
  - Context free grammars from parse tree, stochastic logic programs from proof trees, etc.

Data is labeled:

- $D = \{(\vec{x}_j, \vec{y}_j)\}_{j=1}^{N}$ training set of labeled **input-output** pairs
  - $\vec{x}_j$ … vector of binary, categorical, real valued **features**
  - $\vec{y}_j$ … classes, **labels**

Training and Learning – Training Data

(Real valued) Vectors
- classes (application depended, e.g. flowers, patients, weather forecast)

Text data
- Entities (person names, locations, organisations, ...)
- Sentiments (positive, negative)
- Arguments (pro, cons)

Sequences
- Tumor classification (DNA)

Images
- Objects
- Shapes

Videos
- Gestures

Time Series

...

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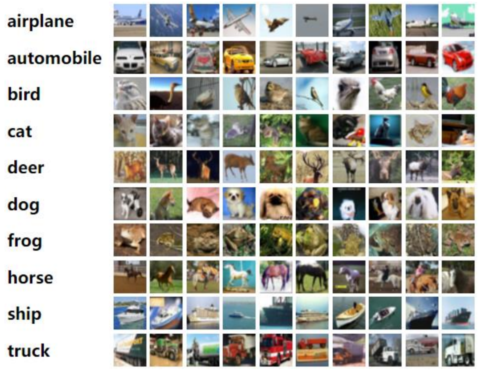
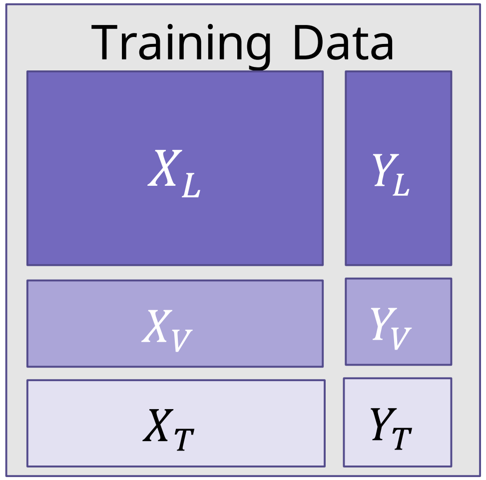
Thisprospectus ORG supplement is not complete without, and may not be delivered orutilized except in connection with, the prospectus, including any amendmentsor supplements to it.There is currently no public market for our common stock. GPE We currentlyintend to have our shares quoted on the OTCQB PERSON operated by the OTC Markets ORG ,although we may not be successful and our shares may never be quoted andowners of our Common Stock PERSON may not have a market in which to sell the shares.Also, no estimate may be given as to the time that this application processwill require.This prospectus supplement incorporates into our prospectus the informationcontained in our Quarterly Report on Form 10-Q for WORK_OF_ART the quarterly DATE period endedSeptember 30, CARDINAL 2017 filed with the Securities and Exchange Commission ORG onNovember 7, 2017 DATE , and our Current Report ORG on Form 8-K filed with the Securitiesand Exchange Commission ORG on November 22, 2017 DATE , each attached hereto.Investing in our common stock involves risks. GPE



https://www.cs.toronto.edu/~kriz/cifar.html

Training and Learning – Training Data

Split $TrainingData = (X, Y)$ (labeled data), into 3 non-overlapping and independent data sets:



Training (or Learning) set $L = (X_L, Y_L)$

a data set where “anything goes . . . including hunches, preliminary testing, looking for patterns, trying large numbers of different models, and eliminating outliers”

Validation set $V = (X_V, Y_V)$

a data set to be used for model selection and assessment of competing models (usually on the basis of predictive ability);

Test set $T = (X_T, Y_T)$

a data set to be used for assessing the performance of a completely specified final model

$TrainingData = \{L, V, T\}$

A.J. Izenman, *Modern Multivariate Statistical Techniques, Springer Texts in Statistics*, Springer, 2008, p.11

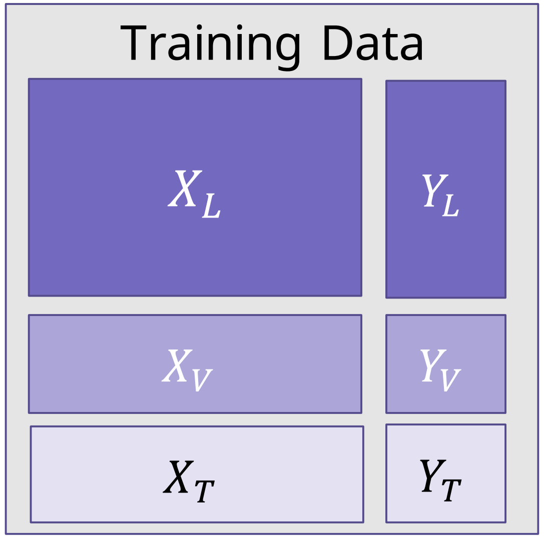
Training and Learning - Process

Task

- Given training data $(\boldsymbol{X},\boldsymbol{Y})$, build a **model** $f(\boldsymbol{X})$ to predict $\boldsymbol{y} \in \boldsymbol{Y}_{\boldsymbol{new}}$ based on unknown $\boldsymbol{x} \in \boldsymbol{X}_{\boldsymbol{new}}$

Strategy

- **Estimate** a function $f\colon \vec{x} \rightarrow \vec{y}$ **on** $(\boldsymbol{X},\boldsymbol{Y})$
- **Hope** that $\boldsymbol{f}(\boldsymbol{x})$ **also works** to **predict unknown** $\boldsymbol{y} \in \boldsymbol{Y}_{\boldsymbol{new}}$ **for** $\boldsymbol{x} \in \boldsymbol{X}_{\boldsymbol{new}}$
- The **“hope”** is called **generalization**

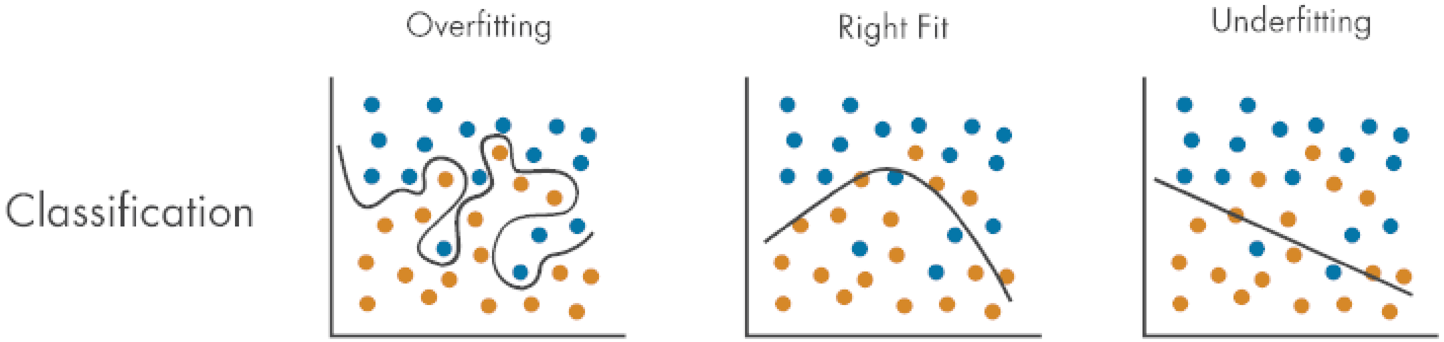


Overfitting

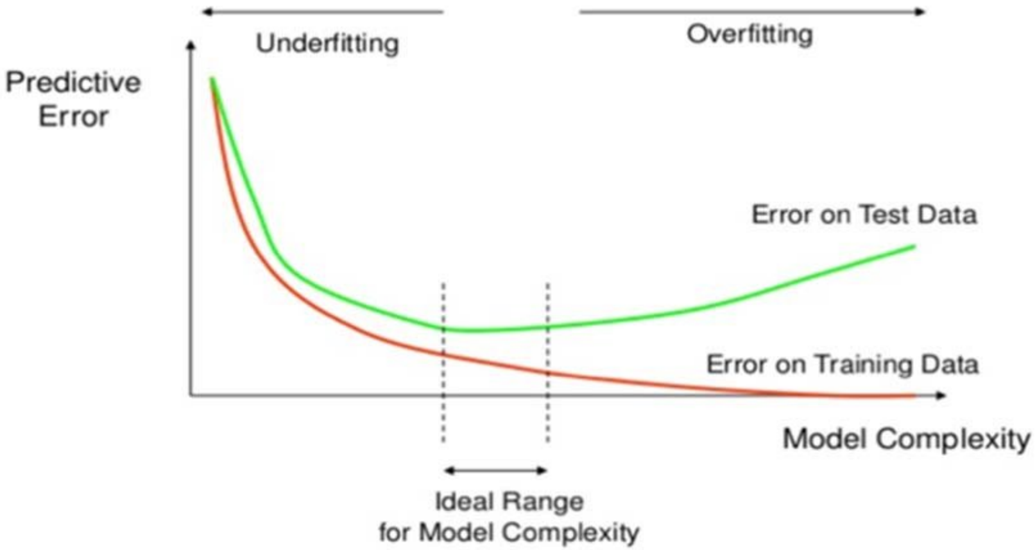
- **Estimated function** *f (x)* **predicts** $\boldsymbol{y} \in \boldsymbol{Y}$ very well but is **unable** to **predict** $\boldsymbol{y}' \in \boldsymbol{Y}_{\boldsymbol{new}}$

We want **to build a model** that **generalizes** well to **unseen data**

Training and Learning - Process



https://www.mathworks.com/discovery/overfitting.html



Al-Behadili, Hayder & Ku-Mahamud, Ku & Sagban, Rafid. (2018). Rule pruning techniques in the ant-miner classification algorithm and its variants: A review

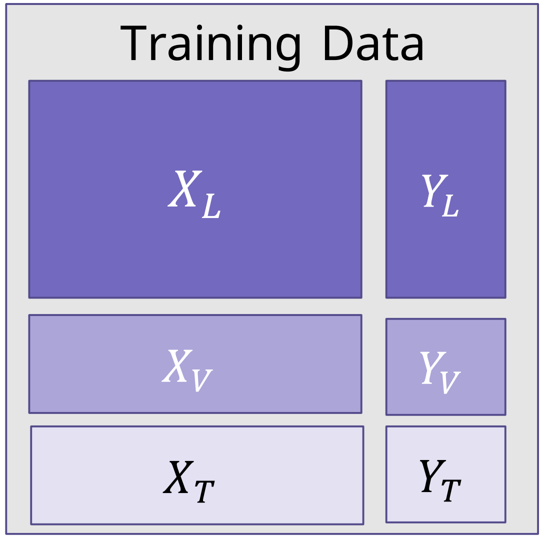
Training and Learning - Process

How to generalize well to unseen data?

Idea:

Pretend not to know data/labels which are **known**

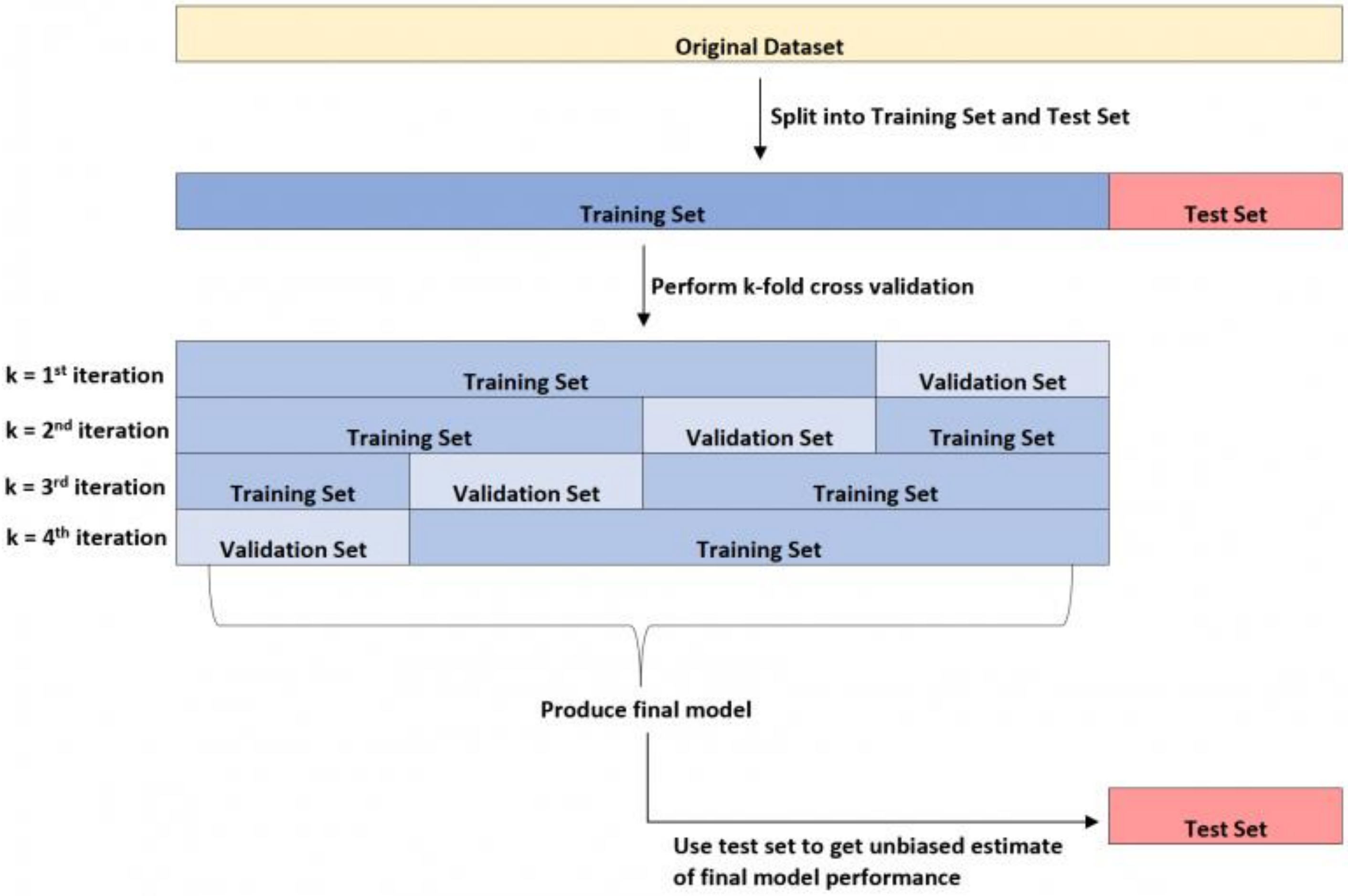
- Build model $f(\boldsymbol{X})$ on the training data
- See how well $f(\boldsymbol{X})$ does on the validation test data
  - If it does well, then apply it also to $\boldsymbol{X_T}$
  - If not, change model



Refinement: Cross validation (10-fold)

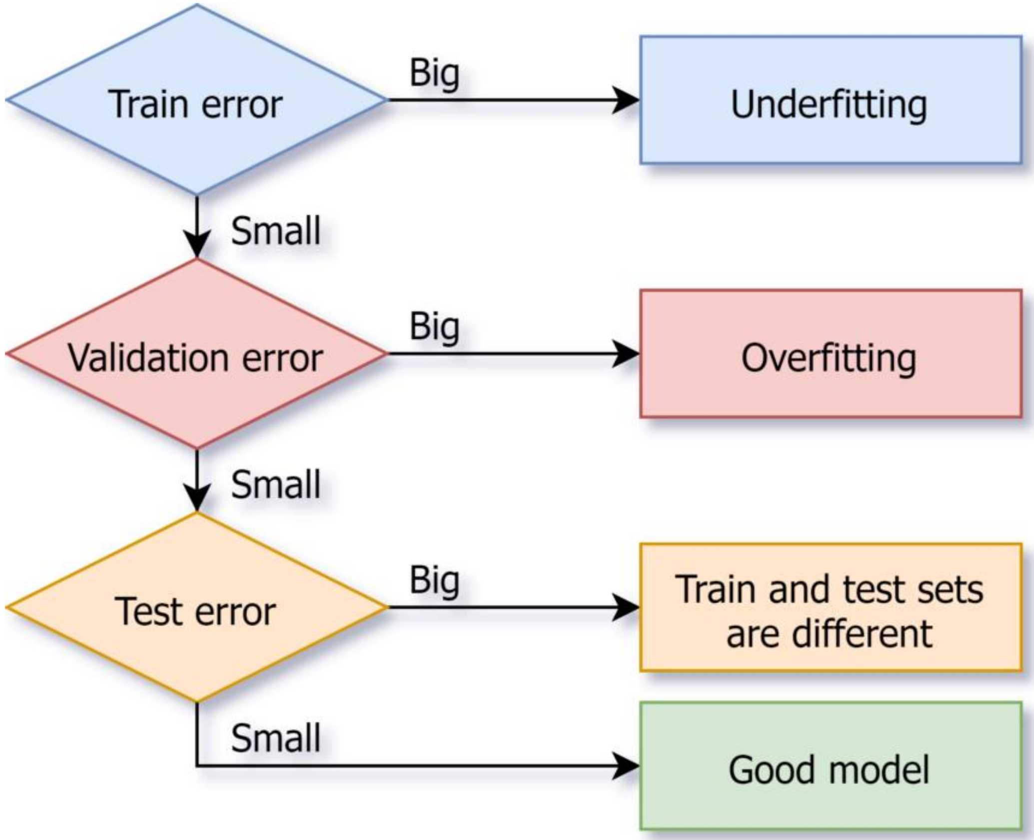
- Splitting into training/validation set is brutal
- Let's split our data ***(X,Y)*** into 10-folds (buckets)
- Take out 1-fold for validation, train on remaining 9
- Repeat this 10 times, report average performance

Training and Learning - Process



https://www.statology.org/validation-set-vs-test-set/

Training and Learning - Errors



https://www.linkedin.com/pulse/overfitting-underfitting-principles-machine-learning-lakshit-

Training and Learning - Inference

Model training

- "Describing" a set of **predetermined classes**
- Each **data point** is assumed to belong to a predefined class, as determined by the class label attribute
- The data used for model training is the training set
- The **trained model** is represented as classification rules, decision trees, mathematical formulae, ...

Model inference - Classifying unknown data

- Validation and test
  - Known label of test sample is compared with the classification result obtained by the model
  - Estimate accuracy of the model
  - Accuracy rate is the percentage of test set samples that are correctly classified by the model
- Classification of unknown data

Training and Learning - Takeaway

Features

Labels

Data

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4.9	3.1	1.5	0.	setosa
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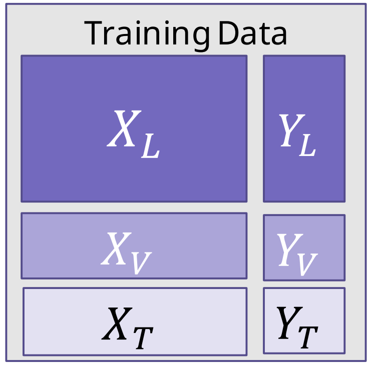
6.0	2.2	5.0	1.5

- Split into train, validation, and test data

Process

- Training and inference are different things



Never test your model on the same data it was trained on!

Structure of this Lecture

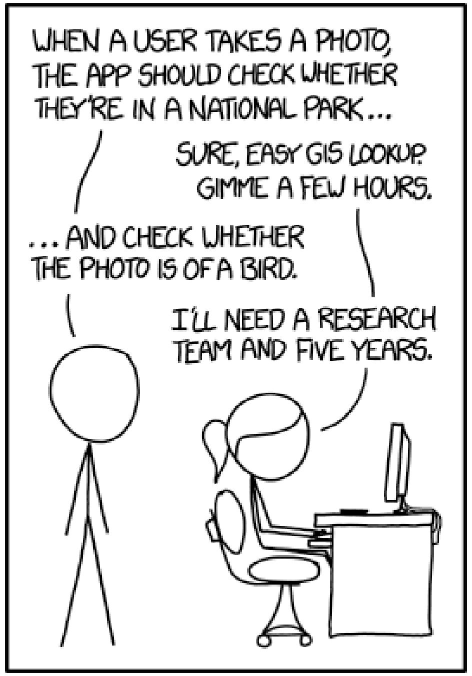
1. Supervised Learning – Training and Learning

2. Classification

    2.1 Decision trees

    2.2 Bayes Classification Methods

- 3. Model Evaluation and Selection



https://xkcd.com/1425/

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Supervised Learning – Decision Tree Example

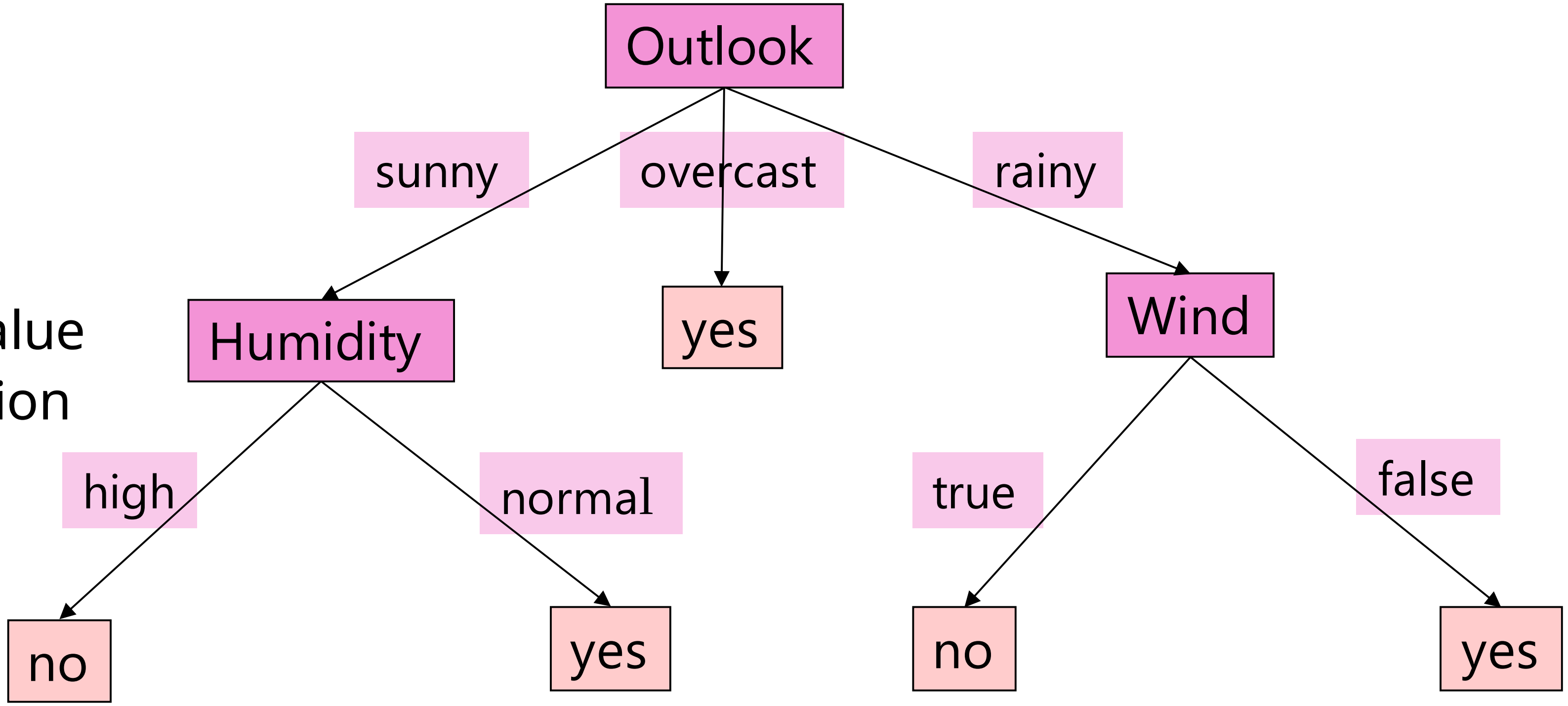
Training Data

Outlook	Temperature	Humidity	Windy	Sport
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	nigh	true	no

Supervised Learning – Decision Tree

Resulting Tree:

- Inner node: feature testing
- Each Branch: One feature value
- Each leaf shows a classification



Supervised Learning – Decision Tree

Goals:

- Unknown data should be correctly classified
- Small tree

Decision Tree - Top-Down Induction

Algorithm (ID3):

Outlook

- **A** := *'best' feature* for decision for subsequent node
- Use **A** as decision for Node
- For each value of **A**, generate new successor of Node
- Sort training data into these leaf nodes
- Training data are perfectly classified

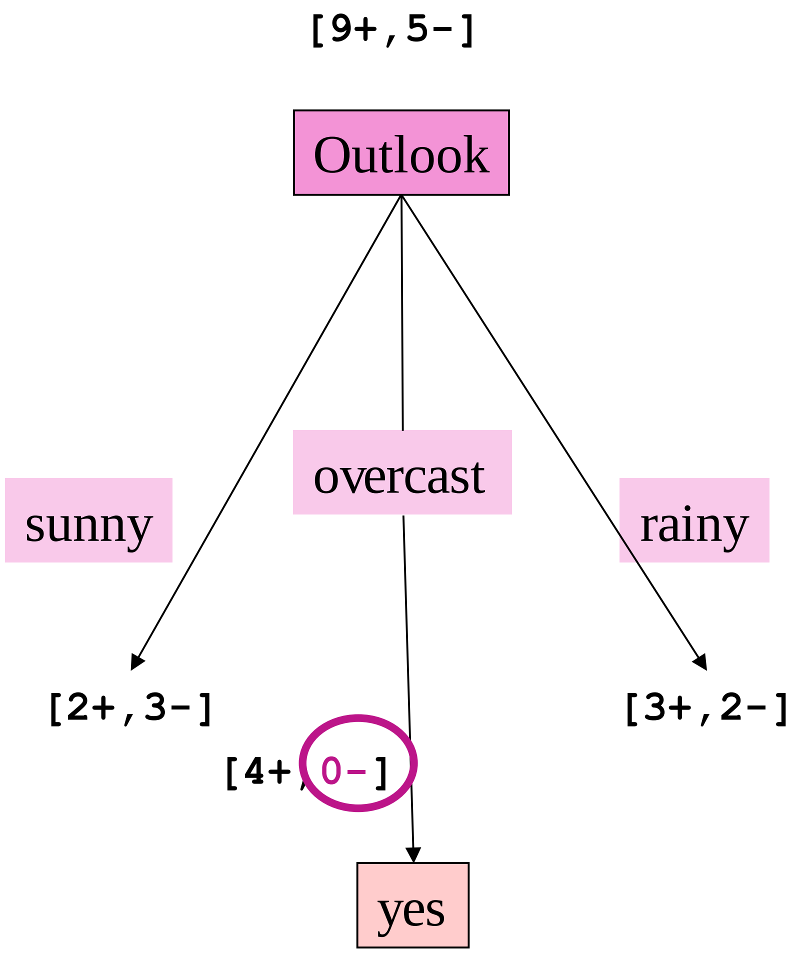
  then STOP,

  else iterate over new leaf nodes

'Best' feature:

Information Gain (see e.g. Han, Kamber, Pei)

Learning a Decision Tree



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Entropy

Entropy H(X) (Greek letter Eta) measures the uncertainty in a variable X. Where x_i are the possible values of X and $p(x_i)$ the corresponding relative frequency. It can be understood as the expected number of bits needed to encode a randomly drawn member of a sample of training examples (under the optimal, shortest-length code).

It is calculated as: $$H(X) = -\sum_{i=1}^{n} p(x_i) \cdot \log_2(p(x_i))$$

Examples:

$$H(X) = -\left(\frac{1}{4} \cdot \log_2 \frac{1}{4} + \frac{1}{4} \cdot \log_2 \frac{1}{4} + \frac{1}{2} \cdot \log_2 \frac{1}{2}\right)$$
$$= -\left(\frac{1}{4} \cdot (-2) + \frac{1}{4} \cdot (-2) + \frac{1}{2} \cdot (-1)\right)$$
$$= 1.5$$

$$H(X) = -(1 \cdot \log_2 1)$$
$$= 0$$

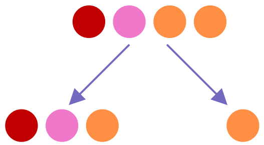
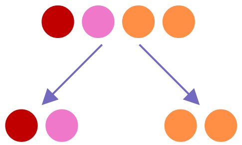
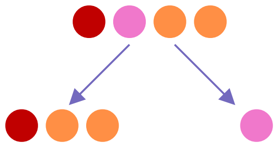
Shannon, Claude Elwood: A Mathematical Theory of Communication.
In: The Bell System Technical Journal, 27 (1948), S. 379–423

Information Gain

Information Gain IG(X,A) measures the expected reduction in entropy H from a prior state to a state where an attribute A has been observed.

It is calculated as: $$\mathrm{IG}(X,A) = H(X) - \sum_{v \in \mathrm{values}(A)} \frac{|X_v|}{|X|} H(X_v)$$

Example: Which split is the best?

		
$\mathrm{IG}(X,A) = 1.5 - \left(\frac{3}{4} \cdot 1.58 + \frac{1}{4} \cdot 0\right)$ = 0.315	$\mathrm{IG}(X,A) = 1.5 - \left(\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0\right)$ = 1.0	$\mathrm{IG}(X,A) = 1.5 - \left(\frac{3}{4} \cdot 0.92 + \frac{1}{4} \cdot 0\right)$ = 0.81

- H = 1.5
- H = $-\log_2 \frac{1}{3} = 1.58$
- H = 0.92
- H = 1
- H = 0
- H = 0

Information Gain – Sports example

$$H(X) = -\sum_{i=1}^{n} p(x_i) \cdot \log_2(p(x_i))$$

$$IG(X, A) = H(X) - \sum_{v \in values(A)} \frac{|X_v|}{|X|} H(X_v)$$

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overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	nigh	true	no

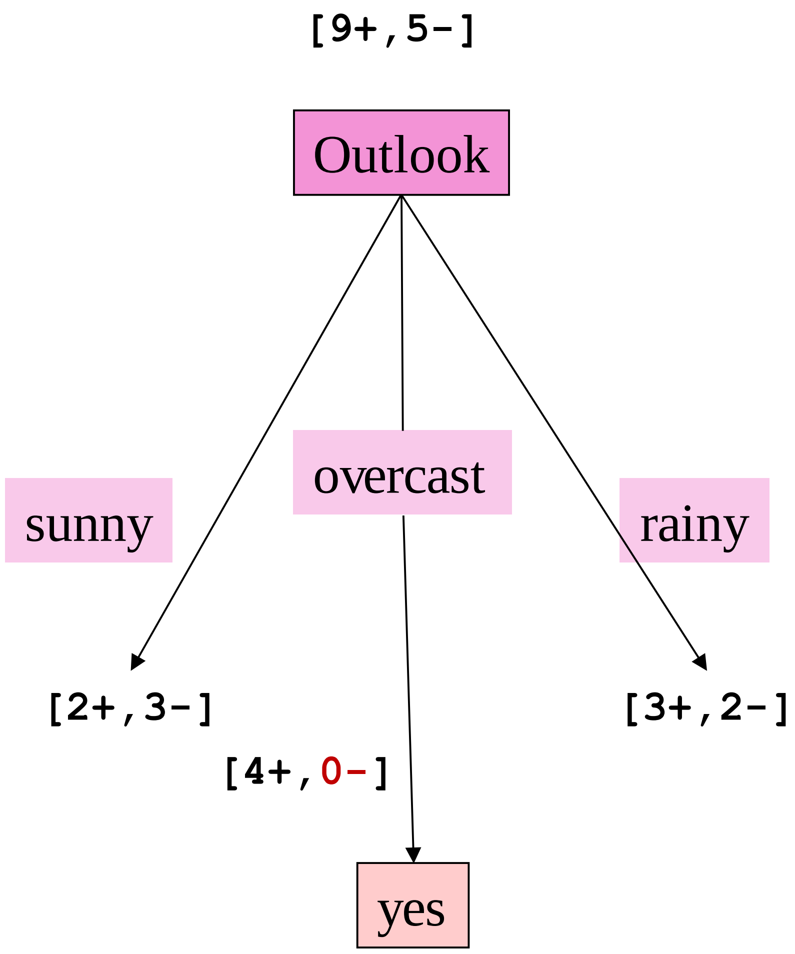
X: Sport

$$H(X) = -\left(\frac{9}{14}\log_2\frac{9}{14} + \frac{5}{14}\log_2\frac{5}{14}\right) \approx 0.94$$

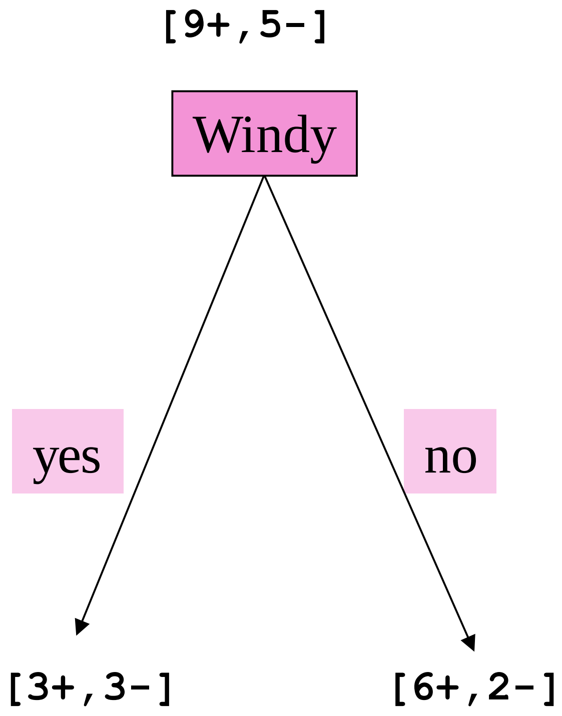
IG(X,Outlook) = 0.94 – $\left(\frac{5}{14}\text{Entropy}(\text{Outlook: sunny}|X) + \frac{4}{14}\text{Entropy}(\text{Outlook: overcast}|X) + \frac{5}{14}\text{Entropy}(\text{Outlook: rainy}|X)\right) \approx \mathbf{0.246}$

IG(X,Windy) = 0.94 – $\left(\frac{6}{14}\text{Entropy}(\text{Windy: true}|X) + \frac{8}{14}\text{Entropy}(\text{Windy: false}|X)\right) \approx \mathbf{0.048}$

Learning a Decision Tree



$IG(Outlook) \approx 0.246$



$IG(Windy) \approx 0.048$

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Supervised Learning – Bayes Classification Methods

A statistical classifier

- performs *probabilistic prediction, i.e.,* **predicts class membership probabilities**

Foundation

- Based on Bayes' Theorem

Performance

- A simple Bayesian classifier, *naïve Bayesian classifier*, has comparable performance with decision tree and selected neural network classifiers

Incremental

- Each training example can incrementally increase/decrease the probability that a hypothesis is correct — prior knowledge can be combined with observed data

Standard

- Even when Bayesian methods are computationally intractable, they can provide a standard of optimal decision making against which other methods can be measured

Bayes’ Theorem: Basics

Example Data

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31 …40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31 …40	low	yes	excellent	yes
…	…	….	…	…

Bayes’ Theorem: Basics

Example Data

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31 …40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31 …40	low	yes	excellent	yes
…	…	…	…	…

Bayes' Theorem: Basics

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31 …40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31 …40	low	yes	excellent	yes

Bayes' Theorem: $P(H|\boldsymbol{X}) = \frac{P(\boldsymbol{X}|H)P(H)}{P(\boldsymbol{X})}$

- Let **X** be a data sample (***evidence***): class label is unknown
  - Example data sample: (<=30, high, no, fair)
- **Classification:**
  - hypothesis H: **X** belongs to class C
  - determine *P(H|**X**)*, i.e., ***posteriori probability***
  - *P(H|**X**):* probability that hypothesis H holds given that we know the attribute description of **X**
  - E.g.: Probability, that (<=30, high, no, fair) buys a computer

Bayes' Theorem: Basics

age	income	student	credit rating	buy computer
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>40	low	yes	excellent	no
31 …40	low	yes	excellent	yes

Bayes' Theorem: $P(H|\boldsymbol{X}) = \dfrac{P(\boldsymbol{X}|H)P(H)}{P(\boldsymbol{X})}$

- **Classification:**
  - determine *P(H|**X**)*, i.e., ***posteriori probability***, hypothesis H: **X** belongs to class C
  - *P(H|**X**):* probability that hypothesis H holds given that we know the attribute description of **X**
  - Probability, that (<=30, high, no, fair) buys a computer

- *P(H)* (***prior probability***): the initial probability
  - E.g., probability that any **X** will buy computer, regardless of age, income, …
- *P(**X**)* (***prior probability***): probability that sample data is observed
  - E.g., probability that (<=30, ?, ?, ?) is an observed data sample
- *P(**X**|H)* (***likelihood***): the probability of observing the sample **X**, given that the hypothesis holds
  - E.g., Given that **X** will buy computer, the probability that **X** is aged between 31… 40, medium income

Prediction Based on Bayes’ Theorem

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31 …40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31 …40	low	yes	excellent	yes

- Given: Training data **X**
- *posteriori probability of a hypothesis* H, *P(H|**X**)*, follows the Bayes’ theorem

$$P(H|\boldsymbol{X}) = \frac{P(\boldsymbol{X}|H)P(H)}{P(\boldsymbol{X})}$$

- Informally: posteriori = likelihood * prior/evidence

Prediction:

- **X** belongs to C_i **iff** the probability $P(C_i|\boldsymbol{X})$ is the highest among all the $P(C_k|\boldsymbol{X})$ for all the *k* classes
- E.g., (<=30, high, no, fair) buys a computer **iff**

  P(**buys a computer**|(<=30, high, no, fair)) > *P*(**does not buy a computer**|(<=30, high, no, fair))

Practical difficulty: requires initial knowledge of many probabilities, involving significant computational cost

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Classification Is to Derive the Maximum Po

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31 ...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31 ...40	low	yes	excellent	yes

- **Given:** Training set D of tuples and associated class labels
- Each tuple is represented by a vector **X** = (x_1, x_2, …, x_n)
- There are *m* classes C_1, C_2, …, C_m

- **Classification: Derive maximum posteriori**, i.e., the maximal $P(C_i|\boldsymbol{X})$
- Can be derived from Bayes' theorem

$$P(C_i|\boldsymbol{X}) = \frac{P(\boldsymbol{X}|C_i)P(C_i)}{P(\boldsymbol{X})}$$

- Since $P(\boldsymbol{X})$ is constant for all classes, only $P(\boldsymbol{X}|C_i)P(C_i)$ needs to be **maximized**
- What about $P(\boldsymbol{X}|C_i)$**?**

Naïve Bayes Classifier

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31 …40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31 …40	low	yes	excellent	yes

Simplified assumption (naïv):

- Attributes are conditionally independent (i.e., no dependence relation between attributes):

$$P(\boldsymbol{X}|C_i)=\prod_{k=1}^{n} P(x_k\,|C_i)= P(x_1\,|C_i)\times P(x_2\,|C_i)\times\cdots\times P(x_n\,|C_i)$$

Greatly reduces costs: only counts the class distribution

- If attribute A_k is categorical:
  - $P(x_k|C_i)$ is the # of tuples in C_i having value x_k for A_k
  - divided by $|C_{i,D}|$ (# of tuples of C_i in training set D)
- If attribute A_k is continous-valued:

  $P(x_k|C_i)$ is usually computed based on Gaussian distribution with mean μ and standard deviation σ

$$P(x_k\,|C_i)=\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Naïve Bayes Classifier - Example

Class:

C_1:buys_computer = 'yes'

C_2:buys_computer = 'no'

Data to be classified:

X = (age <=30,

income = medium,

student = yes

credit_rating = Fair)

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31 …40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31 …40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31 …40	medium	no	excellent	yes
31 …40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayes Classifier - Example

$P(C_i)$: P(buys_computer = “yes”) = 9/14 = 0.643

P(buys_computer = “no”) = 5/14= 0.357

Compute $P(X|C_i)$ for each class:

- P(age = “<=30” | buys_computer = “yes”) = 2/9 = 0.222
- P(age = “<= 30” | buys_computer = “no”) = 3/5 = 0.6
- P(income = “medium” | buys_computer = “yes”) = 4/9 = 0.444
- P(income = “medium” | buys_computer = “no”) = 2/5 = 0.4
- P(student = “yes” | buys_computer = “yes) = 6/9 = 0.667
- P(student = “yes” | buys_computer = “no”) = 1/5 = 0.2
- P(credit_rating = “fair” | buys_computer = “yes”) = 6/9 = 0.667
- P(credit_rating = “fair” | buys_computer = “no”) = 2/5 = 0.4

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31 …40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31 …40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31 …40	medium	no	excellent	yes
31 …40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayes Classifier - Example

$P(C_i)$: P(buys_computer = "yes") = 9/14 = 0.643

P(buys_computer = "no") = 5/14= 0.357

Compute P(X|Ci) for each class:

[...]

X = (age <= 30 , income = medium, student = yes, credit_rating = fair)

$P(X|C_i)$: P(**X**|buys_computer = "yes") = 0.222 x 0.444 x 0.667 x 0.667 = 0.044

P(**X**|buys_computer = "no") = 0.6 x 0.4 x 0.2 x 0.4 = 0.019

$P(X|C_i)*P(C_i)$: P(**X**|buys_computer = "yes") * P(buys_computer = "yes") = 0.028

P(**X**|buys_computer = "no") * P(buys_computer = "no") = 0.007

Therefore, X belongs to class ("buys_computer = yes")

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31 …40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31 …40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31 …40	medium	no	excellent	yes
31 …40	high	yes	fair	yes
>40	medium	no	excellent	no

Avoiding the Zero-Probability Problem

Naïve Bayesian prediction requires

- each conditional probability is **non-zero**
- otherwise, the predicted probability will be zero

$$P(\boldsymbol{X}|C_i) = \prod_{k=1}^{n} P(x_k|C_i)$$

Example:

- dataset with 1000 tuples
- income=low (0), income= medium (990), and income = high (10)

Use **Laplacian correction** (or Laplacian estimator):

- *Adding 1 to each case*
  - Prob(income = low) = 1/1003
  - Prob(income = medium) = 991/1003
  - Prob(income = high) = 11/1003

"Corrected" probabilities estimates are **close to** their **"uncorrected" counterparts**

Naïve Bayes Classifier - Comments

Advantages

- Easy to implement
- Good results obtained in most of the cases

Disadvantages

- Assumption: class conditional independence, therefore loss of accuracy
- Practically, dependencies exist among variables
  - E.g., hospitals:

    patients: profile: age, family history, etc.

    symptoms: fever, cough etc.,

    disease: lung cancer, diabetes, etc.
- Dependencies among these cannot be modeled by Naïve Bayes Classifier

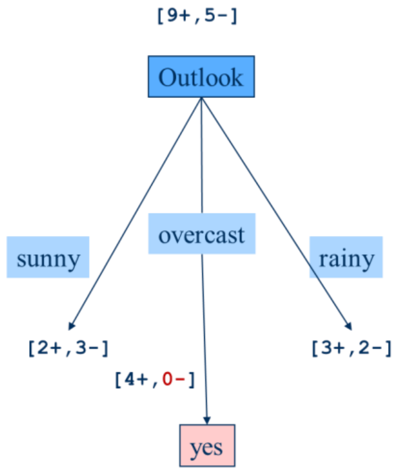
Deal with dependencies

Bayesian Belief Networks https://machinelearningmastery.com/introduction-to-bayesian-belief-networks/

Classification – Other Methods

- Rule-Based Classification
- Support Vector Machines
- k-Nearest-Neighbor Classifier
- Genetic Algorithms
- (Deep) Neural Networks
- ...

Classification - Takeaway



Decision Tree

- Small trees, choose „best“ decisive attribute
- „Best“ is based on entropy and information gain

Naïve Bayes Classifier

- Based on Bayes‘ Theorem
- Precondition attributes are independent!
- Each conditional probability is **non-zero**
- Improvement **Bayesian Belief Networks**

$$P(H|\boldsymbol{X}) = \frac{P(\boldsymbol{X}|H)P(H)}{P(\boldsymbol{X})}$$

$$P(\boldsymbol{X}|C_i) = \prod_{k=1}^{n} P(x_k \mid C_i) = P(x_1 \mid C_i) \times P(x_2 \mid C_i) \times \cdots \times P(x_n \mid C_i)$$

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
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>40	low	yes	fair	yes
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Structure of this Lecture

1. Supervised Learning – Training and Learning
2. Classification

    2.1 Decision trees

    2.2 Bayes Classification Methods

- 3. Model Evaluation and Selection

Model Evaluation and Selection

Use **validation test set** of class-labeled tuples instead of when assessing accuracy

Evaluation metrics:

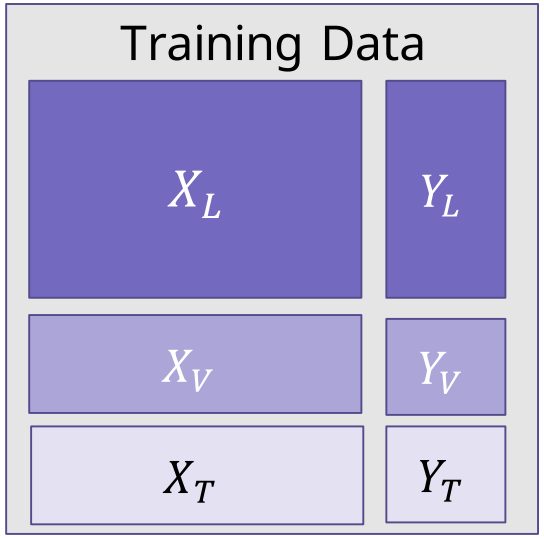
- How can we measure accuracy? Other metrics to consider?

Methods for estimating a classifier's accuracy:

- Holdout method, random subsampling
- Cross-validation
- Bootstrap

Comparing classifiers:

- Confidence intervals
- Cost-benefit analysis and ROC Curves



Classifier Evaluation Metrics: Confusion Matrix

Confusion Matrix:

Actual class \ Predicted class	C_1	¬ C_1
C_1	**True Positives (TP)**	**False Negatives (FN)**
¬ C_1	**False Positives (FP)**	**True Negatives (TN)**

Spamfilter - Example:

Actual class \ Predicted class	is_spam = yes	is_spam = no	Total
is_spam = yes	**6954**	**46**	7000
is_spam = no	**412**	**2588**	3000
Total	7366	2634	10.000

Given *m* classes:

entry C_{ij} in a **confusion matrix** indicates # of tuples in class *i* that were labeled by the classifier as class *j*, $1 \leq i,j \leq m$

- May have extra rows/columns to provide totals

Classifier Evaluation Metrics: Accuracy, Error Rate, Sensitivity, Specificity

A \ P	C	¬C	
C	**TP**	**FN**	**P**
¬C	**FP**	**TN**	**N**
	P'	**N'**	**All**

Classifier Accuracy (Recognition Rate)

- percentage of test set tuples that are correctly classified

Accuracy:

$$Accuracy = \frac{TP + TN}{All}$$

Error rate: *1 – accuracy*, or

$$Error\ rate = \frac{FP + FN}{All}$$

Class Imbalance Problem

- One class may be rare, e.g. fraud, cancer
- Significant majority of the negative class and minority of the positive class
- **Sensitivity** of interest: **True Positive recognition rate**

Sensitivity:

True Positive recognition rate

$$Sensitivity = \frac{TP}{P}$$

Specificity:

True Negative recognition rate

$$Specificity = \frac{TN}{N}$$

Classifier Evaluation Metrics: Precision, Recall, and F-Measures

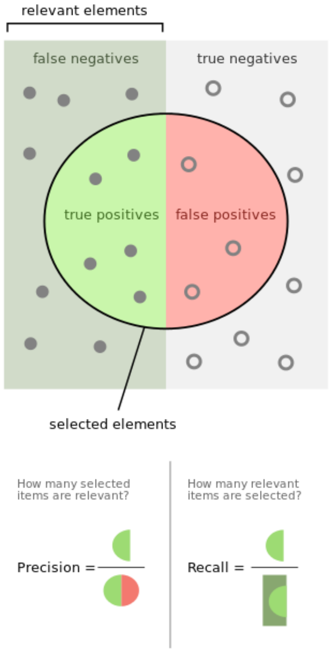
Precision: exactness – what % of tuples that the classifier labeled as positive are actually positive

$$precision = \frac{TP}{TP + FP}$$

Recall: completeness – what % of positive tuples did the classifier label as positive?

$$recall = \frac{TP}{TP + FN}$$

A \ P	C	¬C	
C	**TP**	**FN**	**P**
¬C	**FP**	**TN**	**N**
	P'	**N'**	**All**



Perfect score is 1.0 (precision and recall) – What does it mean?

Often inverse **relationship** between **precision & recall**

F

F_β *measure*

- weighted measure of precision and recall
- assigns ß times as much weight to recall as to precision

$$F_\beta = \frac{(1 + \beta^2) * precision * recall}{\beta^2 * precision + recall}$$

F_1 (or ***F*-score**)

- harmonic mean of precision and recall

$$F_1 = \frac{2 * precision * recall}{precision + recall}$$

Classifier Evaluation Metrics: Example

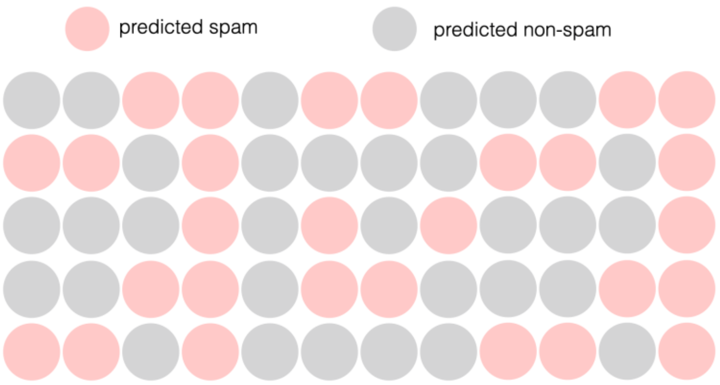
Actual class \ Predicted class	is_spam = yes	is_spam = no	Total	Recognition(%)
is_spam = yes	**6954**	**46**	7000	99.34 (*sensitivity*)
is_spam = no	**412**	**2588**	3000	86.26 (*specificity*)
Total	7366	2634	10.000	95.42 (*accuracy*)

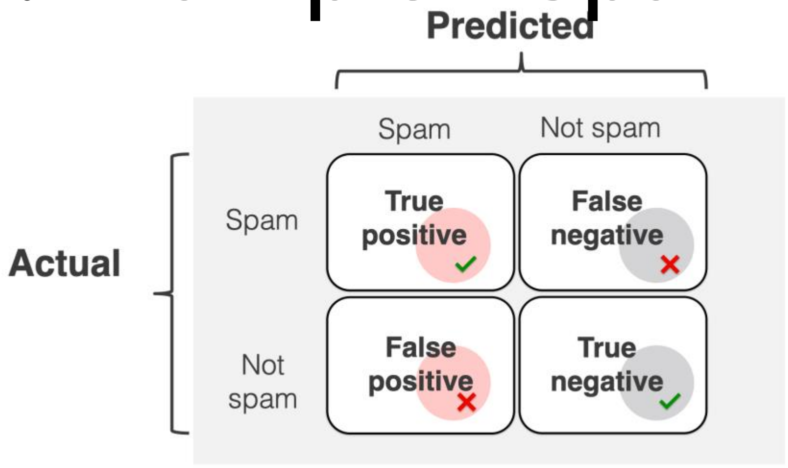
$$Precision = \frac{TP}{TP+FP} = \frac{6954}{7366} = 94.41\%$$

$$Recall = \frac{TP}{TP+FN} = \frac{6954}{7000} = 99.34\%$$

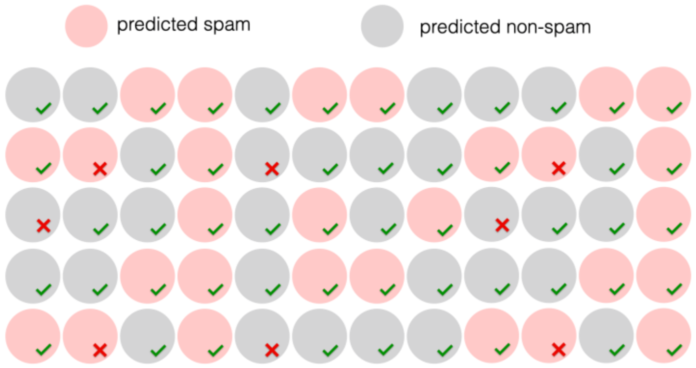
Classifier Evaluation Metrics: Example – Spam Filter 2

Predictions of the Classifier



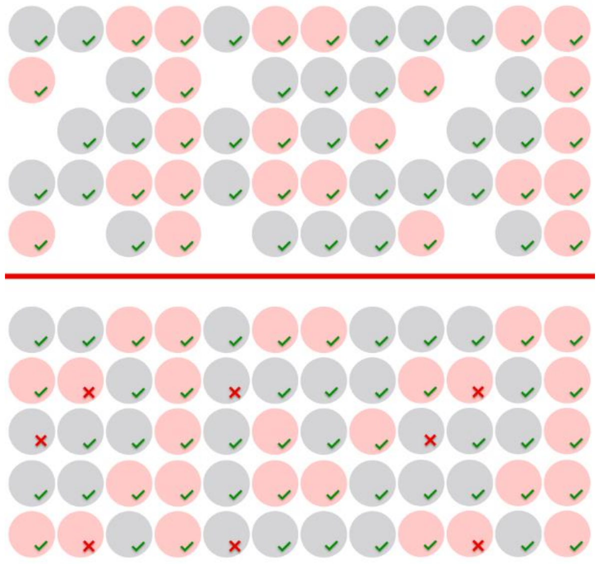


Actual labels



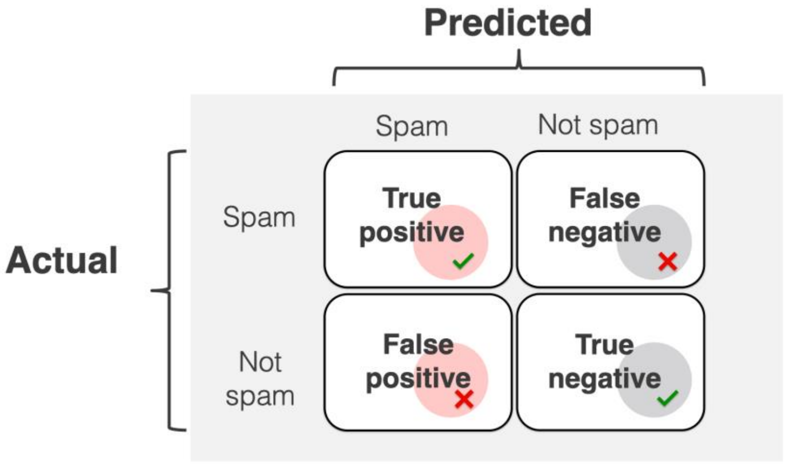
Accuracy

$$Accuracy = \frac{TP+TN}{All} =$$



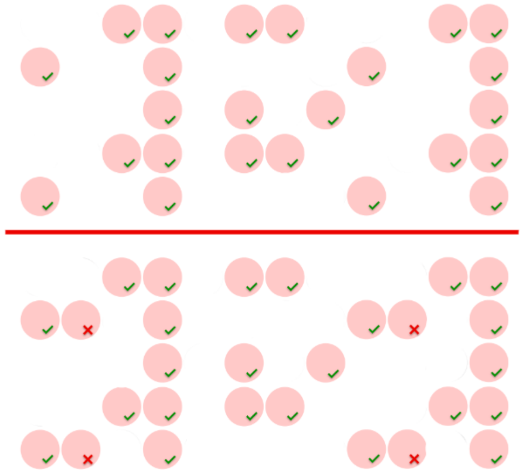
https://www.evidentlyai.com/classification-metrics/accuracy-precision-recall

Classifier Evaluation Metrics: Example – Spam Filter 2

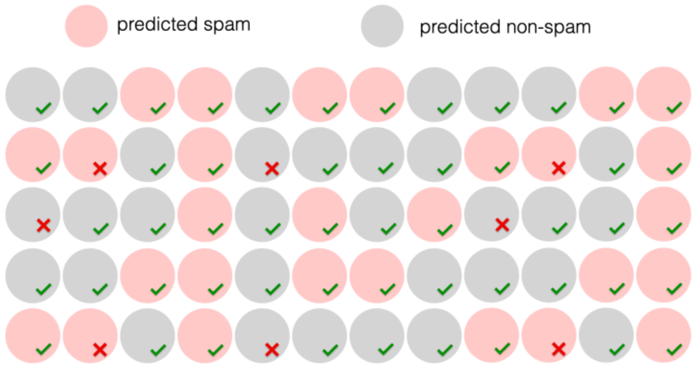


Precision

$$precision = \frac{TP}{TP+FP} =$$

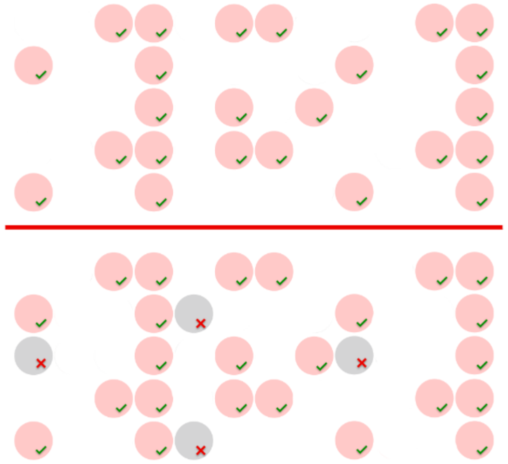


Actual labels



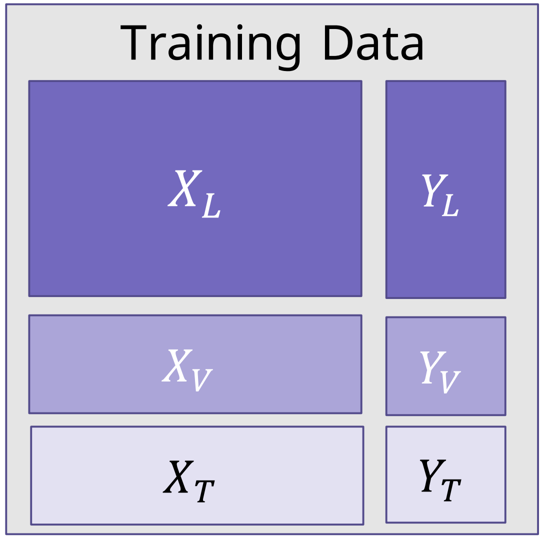
Recall

$$recall = \frac{TP}{TP+FN} =$$



https://www.evidentlyai.com/classification-metrics/accuracy-precision-recall

Evaluating Classifier Accuracy: Holdout and Cross-Validation Methods



Holdout method

- Given data is randomly partitioned into two independent sets
  - Training set (e.g., 2/3) for model construction
  - Test set (e.g., 1/3) for accuracy estimation
- Random sampling: a variation of holdout
  - Repeat holdout k times, accuracy = avg. of the accuracies obtained

Cross-validation (*k*-fold, where k = 10 is most popular)

- Randomly partition the data into *k mutually exclusive* subsets, each approximately equal size
- At *i*-th iteration, use D_i as test set and others as training set
- <u>Leave-one-out</u>: *k* folds where *k* = # of tuples, for small sized data
- <u>***Stratified cross-validation***</u>: folds are stratified so that class distribution in each fold is approximately the same as that in the initial data
  - relatively low bias and variance

Evaluating Classifier Accuracy: Bootstrap

Bootstrap

- Works well with small data sets
- Samples the given training tuples uniformly *with replacement*
  - i.e., each time a tuple is selected, it is equally likely to be selected again and re-added to the training set

The .632 Bootstrap Method

- A data set with *d* tuples is sampled *d* times, with replacement, resulting in a training set of *d* samples
- Data tuples that
- About 63.2% of t … test set (since $(1 - 1/d)^d \approx e^{-1} = 0.368$ if d is large)
- Repeat the sampling procedure *k* times, overall accuracy of the model M is estimated:

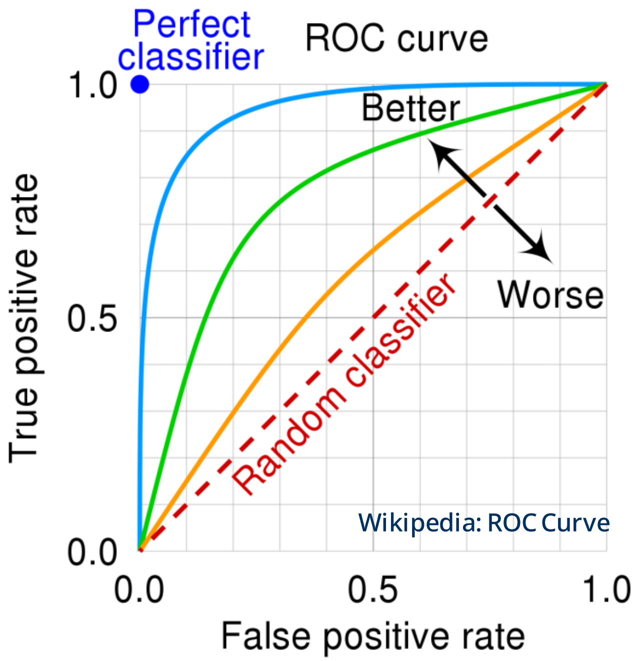
$$Acc(M) = \frac{1}{k}\sum_{i=1}^{k}(0.632 \times Acc(M_i)_{test\_set} + 0.368 \times Acc(M_i)_{train\_set})$$

Model Selection – Statistical Tests

- Given:
  - Two classifiers M_1 and M_2
  - Error rates of M_1 and M_2
- These mean error rates are just ***estimates* of error** on the true population of ***future* data** cases
- What if the difference between the two error rates is just attributed to *chance*?
  - Use a **test of statistical significance** (e.g. t-Test)
  - Be **aware of preconditions** for the respective test
- **Conclude** whether two models are statistically significant different

For details see Han/Kamber 8.5.5

Model Selection: ROC Curves



Wikipedia: ROC Curve

ROC (Receiver Operating Characteristics) curves

- for visual comparison of classification models
- originated from signal detection theory
- vertical axis represents the true positive rate
- horizontal axis represents the false positive rate
- plot also shows a diagonal line

Model M **returns probability** of predicted class for each tuple

Rank test tuples in decreasing order:

the one that is most likely to belong to the positive class appears at the top of the list

- Shows **trade-off** between **true positive rate** and **false positive rate** (1- specifity)
- **Area** under ROC curve is a measure of the accuracy of the model
- The closer to the diagonal line (i.e., the closer the area is to 0.5), the less accurate is the model
- A model with perfect accuracy will have an area of 1.0

Issues Affecting Model Selection

Accuracy

- classifier accuracy: predicting class label

Speed

- time to construct the model (training time)
- time to use the model (classification/prediction time)

Robustness

- handling noise and missing values

Scalability

- efficiency in disk-resident databases

Interpretability

- understanding and insight provided by the model

Other measures, e.g., goodness of rules, such as decision tree size or compactness of classification rules

Machine Learning II

1. Supervised Learning – Training and Learning
2. Classification
    2.1 Decision trees
    2.2 Bayes Classification Methods
- 3. Model Evaluation and Selection

Further Reading

Data Mining

Han, J., Kamber, M., Pei, J.: *Data Mining - Concepts and Techniques*, Morgan Kaufmann, 2012 http://hanj.cs.illinois.edu/bk3/bk3_slidesindex.htm

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Technische Universität Dresden

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Machine Learning II

Digitization and Data Analytics

Deep Learning

Model

Artificial Neural Networks

Task

Classification

Unsupervised Learning

Application Areas

Image Classification

Text Classification

Speech Recognition

…

Deep Learning

Lecture 7

Summary - Machine Learning

Introduction to Machine Learning

Similarity and Distances

Clustering

Outlier detection

Supervised Learning – Training and Learning

Classification

- Decision tree
- Bayes Classification Methods
- Model Evaluation and Selection

Deep learning

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Bayes' Theorem: Basics

Example Da

age	income	student	credit rating	buy computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31 …40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31 …40	low	yes	excellent	yes
…	…	…	…	…

Checken mit A und B, englische A

Information Gain

Information Gain(X, A) = expected reduction in entropy due to sorting on attribute A

$$\boxed{\text{Information Gain(X, A)} = \text{Entropy(X)} - \sum_{v \in values(A)} \frac{X_v}{X} \text{Entropy}(X_v)}$$

Example:

X: sports data

$$\text{Entropy(X)} = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} \approx 0{,}94$$

$$\text{InfoGain(X, Outlook)} = 0{,}94 - \frac{5}{14}\text{Entropy}(Outlook: sunny) - \frac{4}{14}\text{Entropy}(Outlook: overcast) - \frac{5}{14}\text{Entropy}(Outlook: rainy) \approx \mathbf{0,246}$$

$$\text{Information Gain(X, Wind)} = 0{,}94 - \frac{6}{14}\text{Entropy}(Wind: yes) - \frac{8}{14}\text{Entropy}(Wind: no) \approx \mathbf{0.048}$$

References

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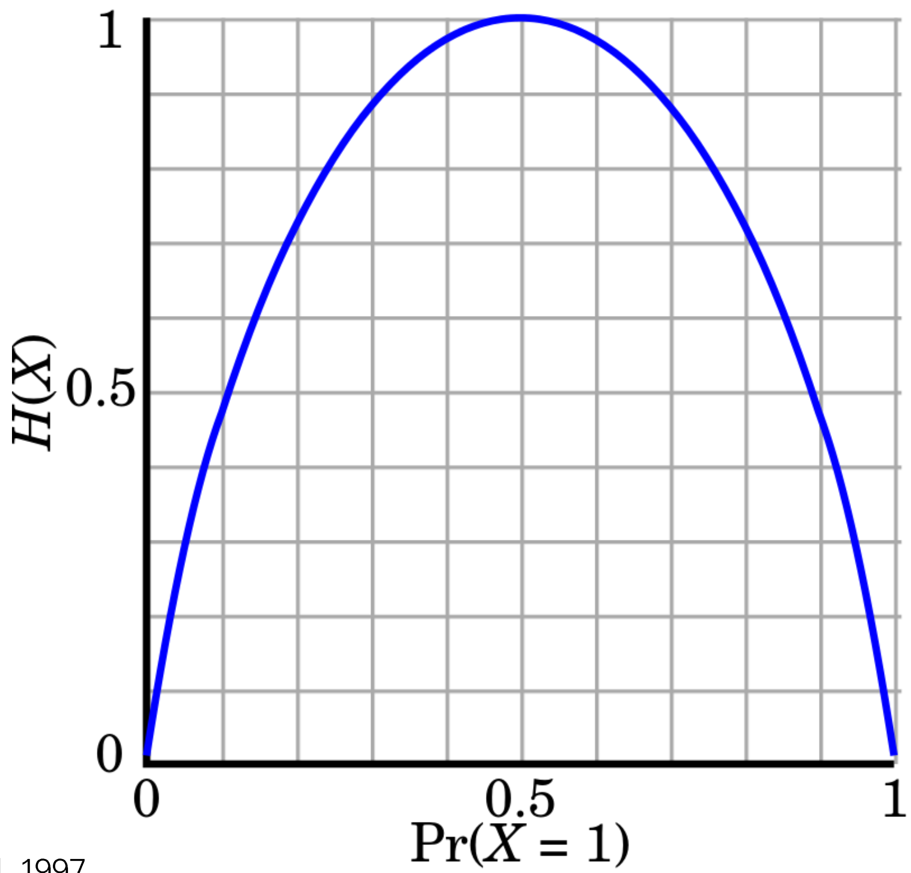
Entropy

- X is a sample of training examples
- $\Pr(X=1) = p_+ = \frac{p}{p+n}$ (the proportion of positive examples in X)
- $\Pr(X=0) = p_- = \frac{n}{p+n}$ (the proportion of negative examples in X)
- Entropy (here $H(X)$) measures the impurity of X

$$Entropy(X) = -p_+ \log_2 p_+ - p_- \log_2 p_-$$



see lecture slides for textbook Machine Learning, c Tom M. Mitchell, McGraw Hill, 1997
http://www.cs.cmu.edu/afs/cs.cmu.edu/project/theo-20/www/mlbook/ch3.pdf

Entropy

Entropy (X) = expected **number of bits** needed **to encode** class (⊕ or ⊖) of randomly drawn member of X (under the optimal, shortest-length code)

Why?

Information theory: optimal length code assigns $-p \log_2 p$ bits to message, which have probability p.

Expected number of bits to encode ⊕ or ⊖ of random member of X:

$$- p_{+} \log_2 p_{+} - p_{-} \log_2 p_{-}$$

Entropy (X) = $- p_{+} \log_2 p_{+} - p_{-} \log_2 p_{-}$

Information Gain

Information Gain(X, A) = expected reduction in entropy due to sorting on attribute A

$$\text{Information Gain(X, A)} = \text{Entropy(X)} - \sum_{v \in values(A)} \frac{|X_v|}{|X|} \text{Entropy}(X_v)$$

Example:

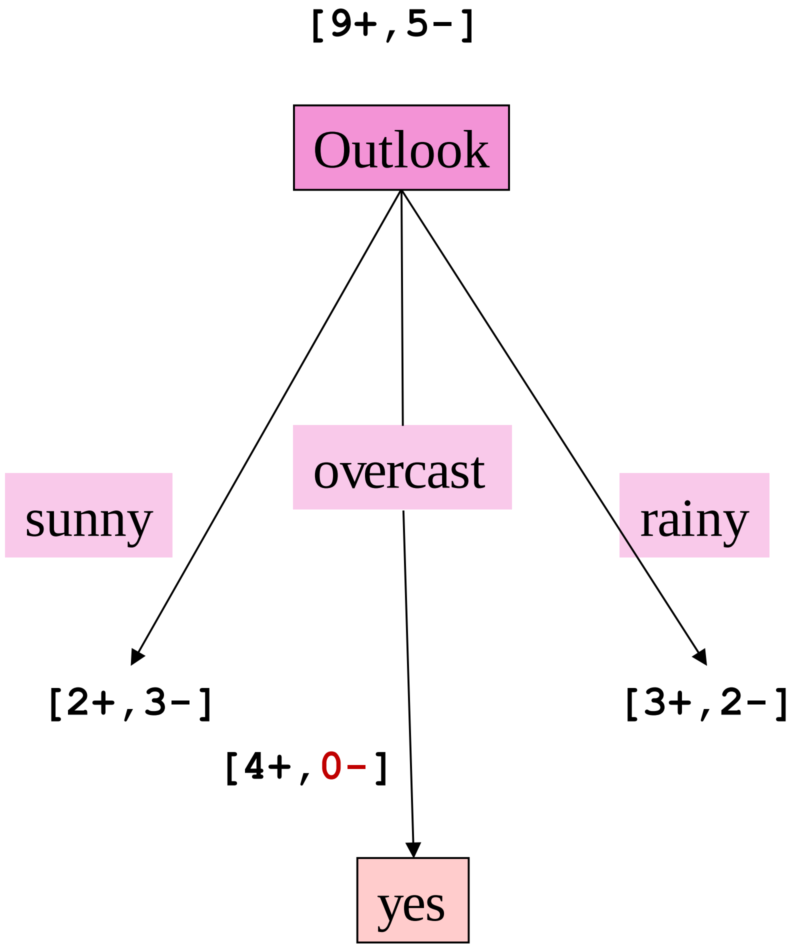
X: sports data

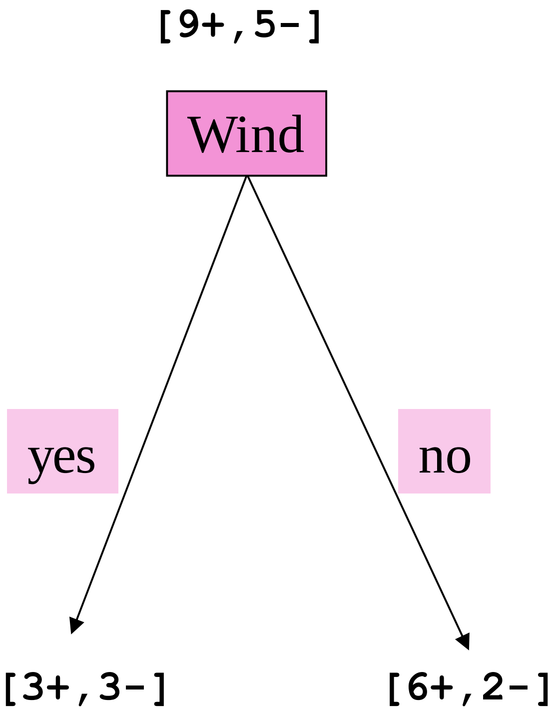
$$\text{Entropy(X)} = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} \approx 0{,}94$$

$$\text{InfoGain(X, Outlook)} = 0{,}94 - \frac{5}{14}\text{Entropy}(Outlook: sunny) - \frac{4}{14}\text{Entropy}(Outlook: overcast) - \frac{5}{14}\text{Entropy}(Outlook: rainy) \approx \mathbf{0{,}246}$$

$$\text{Information Gain(X, Wind)} = 0{,}94 - \frac{6}{14}\text{Entropy}(Wind: yes) - \frac{8}{14}\text{Entropy}(Wind: no) \approx \mathbf{0.048}$$

Learning a Decision Tree





$Information\ Gain(Outlook) \approx 0{,}246$

$Information\ Gain(Wind) \approx 0{,}048$

Random Forest Algorithms

- Ensemble Learning Method
- Trains a large number of decision trees on different randomly chosen subsets of training data
- Each tree being trained on a randomly selected subset of the features

- May handle missing data, noisy data, and irrelevant features
- Less prone to overfitting compared to a single decision tree

- Random forests can be fit in parallel

Classifier Evaluation Metrics: Confusion Matrix

Confusion Matrix:

Actual class \ Predicted class	C_1	$\neg C_1$
C_1	**True Positives (TP)**	**False Negatives (FN)**
$\neg C_1$	**False Positives (FP)**	**True Negatives (TN)**

Example:

Actual class \ Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	**6954**	**46**	7000
buy_computer = no	**412**	**2588**	3000
Total	7366	2634	10.000

Given *m* classes:

entry C_{ij} in a **confusion matrix** indicates # of tuples in class *i* that were labeled by the classifier as class *j*, *1≤i,j≤m*

- May have extra rows/columns to provide totals

Process (1): Model Training

Classification algorithm

Training data

Classifier (model)

IF

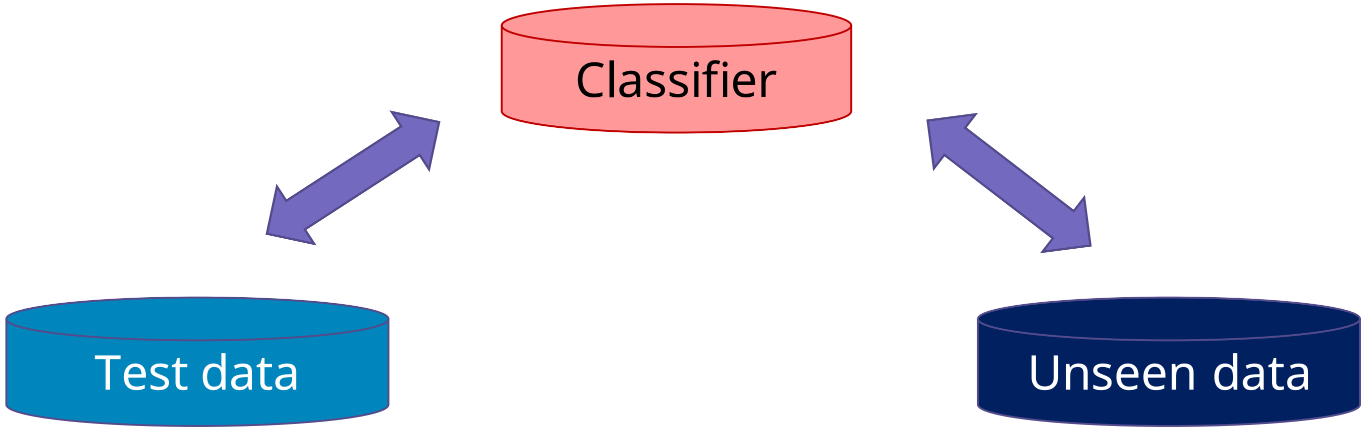
outlook= overcast

THEN sport=yes

Outlook	Temperature	Humidity	Windy	Sport
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes

[J. R. Quinlan: *Induction of Decision Trees.* Mach. Learn. 1, 1.1986]

Process (2): Model Inference - Prediction



Outlook	Temperature	Humidity	Windy	Sport
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes